# The Stubborn Prior:
# When False Scientific Beliefs Distort AI Research Policy

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**Abstract.** If an AI scientist starts with a plausible but wrong theory, does it become confirmation biased? The obvious test compares an agent given a false prior with an agent given a neutral one. That test is confounded: a rational scientist with a false prior should already choose different experiments and end with different beliefs. So we ask a narrower question. Given the belief state the model itself reports, does it do science the way a rational scientist with that same belief state would? We build finite scientific worlds in which every hypothesis, experiment, outcome likelihood, and cost is known, which lets us run an exact Bayesian scientist initialized with the model's own prior. Four completed study families found no detectable effect of a false prior on experiment selection, and a corrected reanalysis of the fixed-evidence arms found that false-prior models moved at least as far against the false theory as neutral ones. That pushed us to a different hypothesis: false priors may leave local reasoning mostly intact while distorting the decisions that control whether contradictory evidence is ever collected. Two proposed studies test whether models avoid a belief-threatening experiment their own prior says to run, and whether they stop investigating when their own prior says to continue. An oracle-only enumeration already shows the first regime exists in useful numbers. A well-powered null is a real result here: it would mean naive false-prior comparisons can overstate apparent bias.

## 1 Introduction

The original question was simple. If an AI scientist starts with the wrong scientific belief, does it become confirmation biased?

It turns out that question does not quite work.

Take two scientists. One thinks theory $H'$ is 80% likely, the other thinks it is 20% likely. Even if both are perfectly rational, they should choose different experiments, value information differently, stop at different times, and finish with different posteriors. Bayes itself predicts a difference. So when a false-prior model behaves differently from a neutral model, that is not evidence of bias. It is what a correct scientist would also do:

$$\text{false prior} \not\Rightarrow \text{irrationality}.$$

A scientist can be wrong without being irrational. A scientist can also reason correctly about each observation and still run bad science, because the theory it holds decides which observations get made. As models move from answering questions to deciding what to test, whether an anomaly is worth chasing, and when a question is done, that second kind of failure is the one that matters.

**The sharper question.** Instead of comparing the false-prior model to a neutral model, we compare it to a rational scientist who believes exactly what the model says it believes:

$$\text{LLM policy} \quad \text{versus} \quad \text{rational policy, same belief state.}$$

The model reports a distribution over hypotheses. We initialize an exact Bayesian agent with that same distribution, in the same world, with the same information, costs, and objective. Any gap between what the model does and what that agent would do is the thing we are studying. False belief on its own is not.

**How the hypothesis moved.** Section 2 reports four completed study families. None found a detectable effect of a false prior on which experiments a model runs. One early result did look like confirmation bias: false-prior models finished with more belief in the false theory than neutral models did, after seeing the same contradictory evidence. That comparison is confounded by the prior itself, and when we measured prior-normalized belief *movement* instead of final posteriors, the false-prior models did not update less against the false theory. So the current evidence does not support "the model sees contradictory evidence and refuses to update."

What it leaves open is a different failure. A model may interpret evidence correctly, and, forced to keep going, may even choose a good next experiment. But it may decide "this is basically settled" before the experiment that would have overturned the theory gets run. If that is right, the failure is not refusing evidence. It is stopping before the evidence arrives. The hypothesis is therefore narrower than the original one: false priors may leave local scientific reasoning relatively intact while distorting the higher-level policy that governs whether inquiry continues. Nothing in the completed studies establishes this. It is what we now propose to test.

Throughout, Section 2 and Section 4.1 report completed work. Everything else is proposed.

## 2 What the Preliminary Studies Actually Showed

Everything in this section is completed. All four families used the same design: a neutral-prior and a false-prior condition (N, F), crossed with whether the model chose its own experiments (E) or received a fixed, replayed evidence stream (Y). The acquisition outcome is the false-minus-neutral difference in wasted information, the information an optimal chooser would have gained but the model did not, in the chooser arms (FE−NE). Every model first had to pass a neutral competence gate. Table 1 gives the results.

**What the table does and does not say.** The manipulations took: every interpretable arm shows a large shift in opening belief toward $H'$. Every acquisition interval crosses zero, and most are wide. Study 1 (Gemma 3 12B, the only one of six open models to pass the competence gate) had neutral identification rate 0.50 and neutral wasted information of 2.88 bits against 3.31 for a random policy, so the baseline itself was weak. Study 2 completed 79/80, 80/80, and 80/80 episodes with in-study neutral competence 0.810, 0.803, and 0.739 for Sol, Grok, and DeepSeek. In study 3, Grok 4.6 shifted its prior by only +0.014 against a prespecified +0.10 threshold and had an acquisition efficiency of 0.113, so its cells say nothing about false-prior effects; Sol's action lists in the two conditions had the same length in 7/8 tasks and the same set in 5/8. Study 4 ran 32/32 episodes with no failures, retries, or exclusions on 7,466 cells, 9 perturbation conditions, and 11 markers. One exploratory result from study 1 is worth recording without overreading: after $H'$ had been refuted, Gemma still reported $H'$ in 0.843 [0.748, 0.927] of remaining false-prior rounds at stated confidence near 0.95. That is a reporting result under a fixed evidence stream, not an acquisition result, and study 2 explains why we do not read it as bias.

### 2.1 The corrected reanalysis

The result that originally looked like confirmation bias came from the fixed-evidence arms. Raw final $p(H')$ gaps between the false-prior and neutral arms shown the same evidence (FX−NY) were +0.128 for Sol, +0.036 for Grok, and +0.092 for DeepSeek in study 2. Those numbers are true and they are not evidence of bias.

Let $H^*$ be the true hypothesis and write the log odds as $\ell_t = \log \frac{p_t(H^*)}{p_t(H')}$. Two agents that start from different priors end at different posteriors even if they process evidence identically. What evidence processing changes is the movement, $U = \ell_T - \ell_0$. For a Bayesian, each observation $y$ from experiment $a$ shifts the pairwise log odds by $\log \frac{P(y|H^*,a)}{P(y|H',a)}$, which does not depend on the prior, on the number of other hypotheses, or on how much mass they carry. So for two arms shown byte-identical evidence, exact Bayes predicts

$$U_{\mathrm{FX}} - U_{\mathrm{NY}} \;=\; 0. \qquad (1)$$

Figure 1 shows the two comparisons side by side. Table 2 gives the corrected estimates. Negative means the false-prior arm moved further against its own favored false hypothesis than the neutral arm did.

Sol in the causal worlds is not distinguishable from the Bayesian zero. Grok and DeepSeek in the causal worlds, and Sol on the real signalling data, moved *further* against the false theory than their neutral counterparts. The existing fixed-evidence data therefore give no support for assimilation shielding.

The Sachs decomposition makes the point exactly. The observed raw FX−NY posterior gap was +0.0146. Exact Bayes, started from the two arms' own elicited opening beliefs and shown the same evidence, predicts +0.0147. Under the untempered oracle the excess residual is about −0.0001, with an interval of about [−0.0003, 0]. The entire apparent persistence was the prior gap carried through the update.

What originally looked like stubborn belief persistence largely disappeared when we asked the correct question. Different priors should produce different final posteriors. Once we measured the evidence-induced movement instead, the false-prior models did not update less against the false theory. That failure of the original result is what motivated the new research question.

## 3 Same-Prior Normative Framework

The same-prior idea is simple to state and easy to get wrong. Four things have to be pinned down: the world, what the model knows, which belief the oracle uses, and how the prior is induced (Figure 2).

### 3.1 The world

A world has a finite hypothesis set $\mathcal{H} = \{h_1, \ldots, h_n\}$, a finite experiment menu $\mathcal{A} = \{a_1, \ldots, a_m\}$, a finite outcome space $\mathcal{Y}$, and known likelihoods $P(y \mid h, a)$ for every

**Table 1.** Completed study families. Intervals are bootstrap 95% intervals. FE−NE is the false-minus-neutral difference in wasted information (bits) in the chooser arms, except study 4, where the outcome is diagnostic value (nats).

| | Model / scale | Prior manipulation | Acquisition, FE−NE | Design lesson |
|---|---|---|---|---|
| 1 | Gemma 3 12B; 48 paired deterministic Boolean worlds | $H'$ adopted at first probe in 0.85 of FE episodes; refuted after a median of 2.0 rounds | +0.105 [−0.471, +0.665]; discriminating-test rate −0.002 [−0.067, +0.063] | Evidence refuted $H'$ within two rounds; no same-prior control |
| 2 | GPT-5.6 Sol, Grok 4.6, DeepSeek V4 Pro; 16 noisy causal worlds, 128 mechanisms, five arms | Opening $p(H')$ shift +0.267, +0.167, +0.352 | Sol +0.123 [−1.172, +1.418]; Grok +0.152 [−0.492, +0.796]; DeepSeek −0.794 [−2.128, +0.540] | Single experiments could settle a world alone; intervals far too wide |
| 3 | GPT-5.6 Sol; 8 computational-lab tasks (Grok 4.6 failed the manipulation gate) | Sol shift +0.140 | +0.018 [−0.025, +0.061]; exact ties in 7/8 tasks | Best and second-best actions nearly equal in value, so choices were uninformative |
| 4 | GPT-5.6 Sol; Sachs single-cell signalling, 8 confirmatory tasks × 4 arms | +0.393 [0.325, 0.451] chooser; +0.453 [0.432, 0.474] replay | +0.021 nats [−0.021, +0.084]; identical measurements in 6/8 tasks | No exact oracle; evidence saturated |

**Table 2.** Corrected fixed-evidence reanalysis. Prior-normalized log-odds movement, false-prior minus neutral arm, under byte-identical evidence. Exact Bayes predicts zero.

| World | Model | $n$ | $U_{\mathrm{FX}} - U_{\mathrm{NY}}$ | 95% interval |
|---|---|---|---|---|
| Causal | GPT-5.6 Sol | 15 | −0.61 | [−1.72, +0.77] |
| Causal | Grok 4.6 | 16 | −0.77 | [−1.36, −0.22] |
| Causal | DeepSeek V4 | 16 | −1.95 | [−2.86, −0.93] |
| Signalling | GPT-5.6 Sol | 8 | −2.75 | [−3.21, −1.96] |

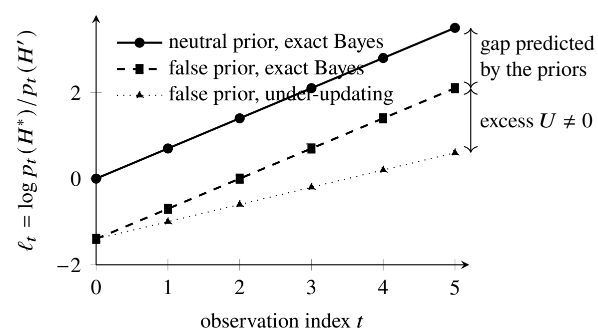


**Figure 1.** Schematic, not data. Two exact Bayesians shown the same evidence move by the same $U$ and end at different posteriors; the final-posterior gap is fully predicted by the prior gap. Only a difference in *movement* (dotted) is evidence about evidence processing. Our early “confirmation bias” result was the upper gap.

combination. One hypothesis $h^* \in \mathcal{H}$ is true. In the false-prior condition the model is induced to favor a specific false hypothesis $H' \neq h^*$. Given a belief $q_t$ and an outcome $y_t$ from experiment $a_t$, the exact update is ordinary Bayes:

$$q_{t+1}(h) = \mathcal{B}(q_t; a_t, y_t)(h) = \frac{q_t(h)\, P(y_t \mid h, a_t)}{\sum_j q_t(h_j)\, P(y_t \mid h_j, a_t)}. \tag{2}$$

Write $P_q(y \mid a) = \sum_h q(h) P(y \mid h, a)$ for the predictive distribution under belief $q$. Because everything is known, the oracle can compute the value of any experiment and of stopping. It knows nothing about the truth that the model does not.

### 3.2 What the model knows

The model and the oracle must face the same information problem. The model is given the hypothesis set, the experiment menu, the outcome space, the likelihood structure, the remaining horizon, experiment costs, and the terminal objective. This makes the task partly computational. That is intentional. We first check that the model can do the computation in a neutral condition (Section 6), and only then ask whether a false prior makes it deviate.

### 3.3 The oracle’s decision problem

Both proposed studies use one decision problem. At a decision with remaining horizon $\tau$ and belief $q$, the agent either stops or runs an experiment $a$ at cost $c(a)$. Stopping

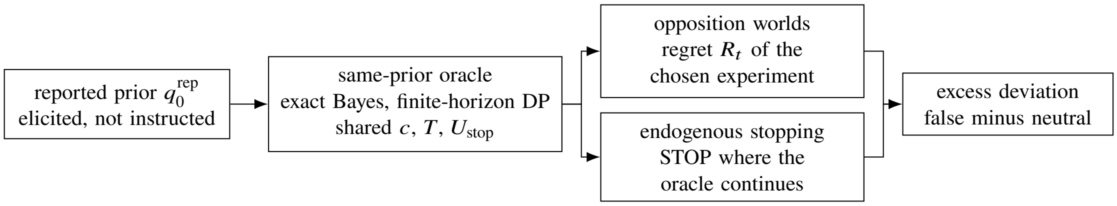


**Figure 2.** The comparison. The oracle is initialized with the model's own reported belief and solves the model's own decision problem. The estimand is never "LLM versus oracle" alone but the false-minus-neutral difference of that gap, so that generic incompetence at the task cancels.

yields terminal utility $U_{\mathrm{stop}}(q)$. Then

$$V_\tau(q) = \max\Big[\, U_{\mathrm{stop}}(q),\ \max_{a\in\mathcal{A}} Q_\tau(q,a)\Big], \tag{3}$$

$$Q_\tau(q,a) = -c(a) + \sum_{y\in\mathcal{Y}} P_q(y \mid a)\, V_{\tau-1}\big(\mathcal{B}(q;a,y)\big), \tag{4}$$

with $V_0(q) = U_{\mathrm{stop}}(q)$. The oracle policy $\pi^*_\tau(q)$ is the maximizer over $\mathcal{A}\cup\{\mathrm{STOP}\}$ (Wald, 1947; DeGroot, 1970).

The terminal utility is a strictly proper scoring rule $S(r,h)$ (Gneiting and Raftery, 2007) applied to a reported final belief $r$ when the truth is $h$. From the agent's own point of view, $U_{\mathrm{stop}}(q) = \max_r \sum_h q(h)\, S(r,h) = \sum_h q(h)\, S(q,h)$, because properness makes honest reporting optimal. Under the log score this is $-\mathrm{H}(q)$ up to a constant, and the one-step continuation value becomes

$$Q_1(q,a) - U_{\mathrm{stop}}(q) = \mathrm{EIG}(a;q) - c(a), \tag{5}$$

the expected information gain of the experiment net of its cost (Lindley, 1956; Howard, 1966). The two proposed studies are the same objective viewed from two decisions: which experiment, and whether any experiment.

**Cost, horizon, and objective are shared.** The model receives exactly the cost function $c$, horizon $T$, and scoring rule $S$ that the oracle optimizes, stated in the prompt. We never tell the model "stop when you are confident" while the oracle optimizes something else. If we did that, we would have built the artifact ourselves.

### 3.4 Which belief the oracle uses

This is central, not a side analysis. A reported belief can differ from the belief a model acts on. If it says $P(H') = 0.6$ but stops as if $P(H') = 0.95$, then "premature closure" might just be misreporting. We keep three belief objects distinct and never substitute one for another silently:

- $q_t^{\mathrm{rep}}$: the distribution the model reports when asked at decision $t$;
- $q_t^{\mathrm{Bayes}}$: the elicited prior $q_0^{\mathrm{rep}}$ propagated by exact Bayes through the evidence the model actually observed;
- $q_t^{\mathrm{rev}}$: a behaviorally inferred belief, defined below and used only in exploratory analysis.

For an exact Bayesian these coincide. For an LLM none of them is guaranteed to equal another, and each estimand below states which one it conditions on.

We run two oracles. The *reported-belief oracle* re-elicits $q_t^{\mathrm{rep}}$ at each decision and compares $a_t^{\mathrm{LLM}}$ with $\pi^*(q_t^{\mathrm{rep}})$: was the action rational given what the model currently says? The *trajectory oracle* compares $a_t^{\mathrm{LLM}}$ with $\pi^*(q_t^{\mathrm{Bayes}})$: does the whole belief-and-action path depart from the rational one? A deviation under the trajectory oracle can come from a belief-updating error, an action error, or both; a deviation under the reported-belief oracle is an action error conditional on the stated belief. We report both and do not net them against each other.

We also check that elicitation is stable across semantically equivalent probability questions (Sclar et al., 2024) and record the dispersion. As an exploratory analysis we fit a one-parameter revealed belief, a scalar shift $\delta$ on the log odds of $H'$,

$$q_t^{\mathrm{rev}}(\delta)(H') \propto q_t^{\mathrm{rep}}(H')\, e^{\delta}, \qquad q_t^{\mathrm{rev}}(\delta)(h) \propto q_t^{\mathrm{rep}}(h),$$

for $h \neq H'$, with $\hat\delta$ chosen per episode so that the oracle policy under $q_t^{\mathrm{rev}}(\hat\delta)$ best matches the model's observed decisions. If the main effect disappears under the revealed belief, the interpretation changes and we say so.

### 3.5 Prior induction

Telling a model $P(H') = 0.8$ does not make it believe that; instructed priors are often ignored (Gupta et al., 2025a). The completed studies induced priors with a memo and checked adoption by elicitation before any comparison. We keep that: the elicited prior $q_0^{\mathrm{rep}}$, not the instructed one, seeds the oracle, and the gap between them is reported. Any effect that survives the primary studies is re-tested with a misleading-pilot induction (Section 7), where the model gets its false belief from honest but unlucky data instead of a memo.

## 4 Opposition Worlds

If an agent strongly favors $H'$, not running the experiment most likely to kill $H'$ can be perfectly rational; positive testing is often a good strategy (Klayman and Ha, 1987). So "false-prior models ran fewer falsifiers" proves nothing.

**Falsifiers, relative to a belief.** Under belief $q$, the falsification probability of $a$ is the agent's own probability that the outcome pushes $H'$ below a threshold $\theta$:

$$F(a;q) = \sum_y P_q(y \mid a)\, \mathbb{1}\big[\mathcal{B}(q;a,y)(H') \le \theta\big]. \tag{6}$$

**Opposition and rational-avoidance worlds.** An *opposition world* is one where the induced prior strongly favors $H'$ ($q_0(H') \ge \pi_{\min}$), $H'$ is false, and under that exact prior the best experiment $a^* = \arg\max_a Q_T(q_0, a)$ is still a falsifier with a value margin over the best non-falsifier:

$$\begin{aligned} F(a^*;q_0) &\ge \phi_{\min}, \\ Q_T(q_0,a^*) - \max_{a:\,F(a;q_0)<\phi_{\min}} Q_T(q_0,a) &\ge g_{\min}. \end{aligned} \tag{7}$$

In these worlds, false prior plus rational policy means run the falsifier. If the model avoids it, its own prior cannot excuse that.

We pair these with *rational-avoidance worlds*, where under the same prior the oracle genuinely prefers a non-falsifying experiment ($F(a^*;q_0) < \phi_{\min}$, with the mirror-image margin). The interesting result is not that models dislike falsifiers. It is that they avoid falsifiers specifically when their own beliefs say they should not. That is an interaction between prior condition and world type, and it separates prior-specific avoidance from a generic habit. World type is defined relative to the false prior; every world is also run under the neutral prior, and each condition's outcome is computed against that condition's own oracle.

### 4.1 Oracle feasibility: opposition states exist in useful numbers

This subsection reports completed oracle-only work. No LLM result is claimed.

The concern with opposition worlds is that they might be rare or pathological, found only by searching after seeing how a model behaves. We checked this by exact enumeration before any model run. The environment has four binary variables $A < B < C < D$ and four candidate causal edges, so $2^4 = 16$ hypotheses; noisy-OR transmission with primary strength $\lambda = 0.8$ and leak 0.10; two readout-sensitivity settings, $(0.9, 0.05)$ primary and $(0.7, 0.20)$; 22 single-readout experiments; 240 ordered $(H^*, H')$ world pairs per noise setting; and false-prior mass $\rho \in \{1/16, 0.25, 0.5, 0.75, 0.9, 0.99\}$. That is 8,640 cells, each computed exactly, with 18 exactness tests passed.

At the primary noise setting and $\rho \ge 0.5$, the falsifier belongs to the false-prior Bayesian top-EIG set in 131 of 240 worlds (55%) and is uniquely optimal in 67 of 240 (28%). In 29 worlds the false prior makes the falsifier preferred when the neutral prior did not; in 52 it removes the falsifier from the top set, which is the rational-avoidance regime. At $\rho = 0.9$ the falsifier ranks first in 125 worlds, second in 41, third in 41, and never worse than tenth of 22.

So the opposition experiment does not depend on finding a pathological world after looking at model behavior. Diagnostically useful states exist in a small, ordinary environment before a target LLM is run. The thresholds $\pi_{\min}$, $\theta$, $\phi_{\min}$, $g_{\min}$ for the frozen world set are chosen from this enumeration (Section 6).

**Primary outcome: value regret.** The primary outcome is the value lost by the model's choice, computed under the model's belief $q_t \in \{q_t^{\text{rep}}, q_t^{\text{Bayes}}\}$:

$$R_t = Q_\tau\big(q_t, \pi^*(q_t)\big) - Q_\tau\big(q_t, a_t^{\text{LLM}}\big). \tag{8}$$

Binary "picked the falsifier" is secondary. Regret is the direct fix for study 3, where a model could pick the second-best action at almost no cost and still get labelled wrong. Regret is in units of the shared utility, so it is comparable across conditions, but the range of attainable regret depends on the belief state; we also report regret normalized by the oracle's best-minus-worst gap as a sensitivity analysis. In this study the model must choose an experiment (STOP is not in the action set); the primary decision is the constructed opposition state at $t = 0$, with later visited states classified by the same criterion as a secondary analysis.

**Estimand.** Since the oracle has zero regret against itself, the prior-specific estimand in world type $w \in \{\text{opp}, \text{RA}\}$ is

$$\Delta^w_{\text{opp}} = \mathbb{E}[R \mid \text{false}, w] - \mathbb{E}[R \mid \text{neutral}, w], \tag{9}$$

and the primary test is the interaction $\Delta^{\text{opp}}_{\text{opp}} - \Delta^{\text{RA}}_{\text{opp}}$. A model that is simply bad at the task has positive regret in both conditions and both world types, and drops out of this contrast.

## 5 Endogenous Stopping

This is the study with the most upside. Most agent benchmarks force the model to keep acting. A real scientist decides when to stop. So the action set is $\mathcal{A} \cup \{\text{STOP}\}$, and the oracle solves Equations (3)–(4). The oracle tells us whether another experiment is worth its cost. The event we care about is the model choosing STOP in a state where the same-prior oracle says CONTINUE. We call that *premature epistemic closure*.

**The estimand is a difference-in-differences.** LLMs may be bad at optimal stopping in general, so the raw rate of stopping-when-oracle-continues is not the result. For condition $k \in \{\text{false}, \text{neutral}\}$ let

$$C_k = P\big(a_t^{\text{LLM}} = \text{STOP} \,\big|\, \pi^*(q_t) \neq \text{STOP},\ k\big), \tag{10}$$

with $q_t$ again $q_t^{\text{rep}}$ (primary) or $q_t^{\text{Bayes}}$. The prior-specific effect is

$$\Delta_{\text{closure}} = C_{\text{false}} - C_{\text{neutral}}. \tag{11}$$

$C_{\text{neutral}}$ measures how badly the model stops when nothing is at stake for a favored theory; $\Delta_{\text{closure}}$ measures how much worse it gets when something is. We also report the reverse error, continuing where the oracle stops, in both conditions. A model that never stops has $C_k = 0$ and is not thereby rational; the reverse error keeps that visible.

**State matching.** The CONTINUE states under a false prior and under a neutral prior are different sets of belief states, and they can sit at different distances from the stopping boundary. Comparing them raw can produce a spurious effect. We therefore define the stopping margin

$$L_{\text{stop}}(q) = \max_a Q_\tau(q, a) - U_{\text{stop}}(q), \tag{12}$$

the value destroyed by stopping in that state, and include it as a covariate in the primary model. We also restrict the headline estimate to states with $L_{\text{stop}} \geq \lambda$ for a prespecified $\lambda$, each condition's margin computed under its own belief and $\lambda$ chosen from oracle simulations so that both conditions have adequate support. This fixes the state-matching problem and separates a meaningful closure effect from one that only appears where continuing barely matters. We report $\Delta_{\text{closure}}(\lambda)$ across a grid of $\lambda$, and a value-weighted version, the expected $L_{\text{stop}}$ destroyed by stopping in oracle-CONTINUE states, in each condition.

**Cost and utility are not hidden.** The stopping boundary depends on $c(a)$ and $U_{\text{stop}}$. We fix one primary regime $(c, S, T)$ in advance, give it to the model, and then rerun across a prespecified range of costs and scoring-rule parameters. A real closure effect should not live at one hand-picked threshold.

## 6 Experimental Engineering and Validation

The biggest lesson from the completed studies is that we kept finding out the environment was wrong after the expensive runs. The new pipeline is

$$\text{mechanism} \to \text{oracle} \to \text{diagnostic regime}$$
$$\to \text{synthetic validation} \to \text{freeze} \to \text{frontier models},$$

and no frontier-model call happens before the first four steps pass. The oracle enumeration of Section 4.1 is the second step for the opposition study.

**Oracle-only validation.** Before any LLM call we generate worlds, compute exact updates, find opposition states and STOP/CONTINUE regions, and run the whole analysis on agents we control: an exact Bayesian agent; a *falsifier-averse* agent acting on $\tilde{Q}(q, a) = Q_\tau(q, a) - \kappa F(a; q)$ for a known $\kappa > 0$; and a *premature-closure* agent acting on $\tilde{U}_{\text{stop}}(q) = U_{\text{stop}}(q) + \beta \mathbb{1}[\arg\max_h q(h) = H']$ for a known $\beta > 0$, a bonus to STOP when its favored hypothesis is ahead. If we implant a 10 percentage-point closure effect, the pipeline should recover roughly 10 points with useful precision. If it cannot, the benchmark is not ready and we fix it before spending anything on models. This is also what makes a later null mean something.

**World quality gates.** A world enters model evaluation only if it satisfies constraints fixed in advance: moderate per-experiment evidence (a cap on likelihood ratios), a meaningful gap between best and second-best action value, a recoverable truth within the horizon, both real CONTINUE and real STOP states, opposition states under the false prior with matching neutral-prior controls, and no single experiment that dominates everywhere. Thresholds come from oracle simulations, not from model behavior. We log the rejection rate; if almost every random world fails, we redesign the generator rather than lean on extreme post-selection.

**Freeze.** World-selection rules, quality thresholds, metrics, and analysis code are committed before any target-model run, as they were in the completed studies. Selection may depend on oracle geometry. It may not depend on whether a particular model behaved interestingly.

**Neutral competence gate.** We do not measure subtle prior-specific bias on a task the model fundamentally cannot solve. Before any false-prior cell, the model is run in the neutral condition; the working target is roughly 85 to 90% oracle-optimal action choice in the relevant states, with low regret. The number is not sacred; the principle is. If competence is poor we may fix task complexity, prompt, interface, or world difficulty, for a small fixed number of revision rounds set before the first run, all logged, each re-checked with the synthetic agents. If competence is still poor, the model is recorded as not competent on this task and excluded from strong claims rather than tuned further, as Grok 4.6 was in study 3.

## 7 Robustness and Statistical Plan

**Prompts.** At the pilot stage we run about five semantically equivalent formulations with identical information, cost, and objective. An effect that vanishes under rewording is not a mechanism, and we want to learn that cheaply.

**Models.** Final evaluation should cover several model families; the completed studies used OpenAI GPT-5.6 Sol, xAI Grok 4.6, DeepSeek V4 Pro, and Google Gemma 3 12B. The early signal that matters is not identical effect size but whether the direction of prior-specific deviation is the same across families.

**Dose response.** For any serious positive we vary prior strength, for example instructed $q_0(H') \in \{0.5, 0.7, 0.9\}$, analyze against the elicited prior in each cell, and ask whether excess deviation from the same-prior oracle increases monotonically. A curve is much stronger evidence than a single treatment-control gap.

**Misleading-pilot induction.** The obvious objection to memo-induced priors is that the model is just following instructions. So any surviving effect is re-tested with priors the model earned honestly. Pilot data are drawn from the true hypothesis, $D_{\text{pilot}} \sim P(\cdot \mid h^*, a_{\text{pilot}})$, and we select finite samples for which a neutral-prior Bayesian posterior legitimately favors $H'$. The model is shown the pilot, its post-pilot belief is elicited, and that elicited belief initializes the oracle. The model becomes wrong for good reasons. If the effect reproduces, instruction-following is no longer a plausible explanation.

**Statistics and power.** We care about effect sizes and intervals, not isolated p-values. For stopping, the primary model is a hierarchical logistic regression of the STOP indicator on prior condition, oracle recommendation, their interaction, $L_{\text{stop}}$, and random effects for model, world, and prompt variant; the coefficient of interest is the interaction, reported as a marginal effect on the probability scale. For regret and value loss we use hierarchical continuous models or bootstrap intervals across worlds. Before any model run we simulate the full analysis with the synthetic agents and pick the per-cell $N$ that detects the smallest effect we would care about: provisionally 10 percentage points of excess closure above the margin threshold, with 300 episodes per model per principal condition, to be revised by the power simulation and the budget. The completed studies' intervals show why this matters: 16 worlds per model gave acquisition intervals more than two bits wide. The full matrix is written down before final runs; if it is too expensive, we cut prompt variants and replicates before model families.

## 8 Related Work

Existing work separately studies Bayesian information gathering, AI scientific agents, belief updating, and confirmation-like test selection. Our question is whether a sequential scientific policy is rational conditional on the model's own belief state, under a controlled false prior. We compare against the nearest work in each of the four literatures.

**Bayesian experimental design and information seeking.** The normative machinery is classical: Lindley (1956) defines the expected information of an experiment, Chaloner and Verdinelli (1995) put design choice inside expected-utility maximization, which is what makes "optimal" depend on the utility, Rainforth et al. (2024) and Foster et al. (2021) cover the modern sequential versions, and Cheng and Huan (2025) treat the coupled design-and-stopping problem without LLMs. The closest methodological neighbor is BED-LLM (Choudhury et al., 2026), which builds a probabilistic model from LLM-sampled hypotheses and LLM likelihoods, chooses each query by expected information gain, and filters hypotheses incompatible with the history. Its motivation already notes that LLM priors are "heavily overconfident on a small number of possible hypotheses." The difference is what the Bayesian policy is for. BED-LLM uses it as a method to make the LLM a better asker and evaluates task success against prompting baselines, with fixed question budgets and no cost-aware stopping. We use an exact same-belief policy as a control, inject false priors as a treatment, and measure regret and stop/continue disagreement against it. Grand et al. (2026) have an explicit ask-versus-act rule and an EIG-based reference, but the reference is computed under the system's own particle posterior rather than the model's stated belief, so it cannot separate false belief from policy distortion. BoxingGym (Gandhi et al., 2025) scores LLM designs by information regret against the environment's true model. Neither conditions on what the model believes.

**Autonomous AI scientists and experimental-design agents.** End-to-end systems generate ideas, run experiments, and write papers (Lu et al., 2024; Yamada et al., 2025) and drive laboratory hardware (Boiko et al., 2023). They are evaluated on output quality or task success, which leaves the inquiry policy unexamined: AI Scientist-v2's tree search over experiments is a research policy, and it is scored only through the resulting paper. Ríos-García et al. (2026) find, across many agent traces, that evidence is frequently ignored and refutation-driven revision is rare. AutoDiscovery (Agarwal et al., 2025) elicits the LLM's own prior and posterior over hypotheses but uses their difference as a search reward, not as a standard for judging the agent. Interactive benchmarks such as DiscoveryWorld (Jansen et al., 2024) and PhysGym (Chen et al., 2025) exercise closed-loop experiment selection but score outcomes (task completion, actions taken, knowledge discovered) rather than the rationality of each choice given the agent's beliefs, and PhysGym varies the prior knowledge *given* to the agent rather than the prior the agent holds. Gupta et al. (2025b) report that LLM agents used directly as optimizers show no sensitivity to experimental feedback, a policy-level failure of the kind we want to measure, but one measured without a belief state.

**Belief updating and stated versus revealed beliefs.** Our oracle is initialized from a verbalized probability distribution, so it inherits what is known about verbalized calibration (Tian et al., 2023; Xiong et al., 2024), including overconfidence and dependence on how the question is asked (Sclar et al., 2024). The stronger concern is that the belief a model states is not the belief it acts on. Yamin et al. (2026) jointly elicit probabilities and decisions and derive utility-free conditions for the stated belief to rationalize the decision; Pal et al. (2025) find models betting against their own high-confidence answers and failing to seek information when confidence is low. Our two-oracle design and the revealed-belief fit $\hat{\delta}$ are our engagement with this measurement problem, applied to sequential experiment choice and stopping rather than to one-shot decisions. On updating itself, Gupta et al. (2025a) find that instructed priors are often ignored while subsequent updates track Bayes, the precedent for our instructed-versus-elicited check. Chen et al. (2026) elicit an LLM's own prior and likelihood and compare its stated posterior with the exact Bayes posterior computed from those same quantities; that is the same-belief-control idea applied to passive updating, and our extension is from inference to policy.

**Confirmation bias, falsification, and sycophancy.** Closest to our opposition experiment, Jhaveri et al. (2026) port Wason's 2-4-6 task (Wason, 1960) to LLMs and measure confirmation bias as the ratio of hypothesis-incompatible to hypothesis-compatible tests relative to the model's own stated hypothesis; FalsifyBench (Bertolazzi et al., 2026) does likewise with semantic rules. Both observe the phenomenon we care about, with no manipulated prior and no Bayesian reference policy, so a preference for compatible tests cannot be told apart from a rational positive test strategy. That distinction is old: Klayman and Ha (1987) showed positive testing is often a good heuristic, and Oaksford and Chater (1994) showed selection-task choices look rational as optimal data selection. Both are the reason our outcome is value regret against an environment-specific oracle rather than a count of falsifiers. Nickerson (1998) separates search-level from interpretation-level bias, the split between our fixed-evidence preliminary studies and the two primary experiments. Sycophancy is agreement with a user's stated view (Sharma et al., 2024), and belief persistence under conflicting documents (Xie et al., 2024) is a related phenomenon; Atwell et al. (2025) score sycophantic belief shifts as deviations from Bayesian consistency, the nearest of these to our framing, but the outcome is a belief and the perturbation is user input. Batista and Griffiths (2026) give a rational analysis in which a Bayesian fed data conditioned on its own hypothesis grows confident without approaching the truth; the agents there are humans, but the mechanism is the one we test in the LLM's own policy. In all of these the outcome is agreement or textual persistence. Ours is a scientific action relative to a same-prior normative policy, with no user opinion in play. We did not find work that combines a planted false prior, an exact Bayesian policy seeded with the model's own reported prior, and outcomes on experiment choice and stopping, and we claim no more than that.

## 9 Limitations and Decision Rules

Each line is a way the result could be wrong, then the control already in the design.

- **Stated beliefs may not be decision-relevant beliefs.** Two oracles; elicitation stability across paraphrases; the revealed-belief fit $\hat{\delta}$; if the effect survives only under the stated belief, that is the finding.
- **Generic stopping incompetence can mimic closure.** Neutral competence gate; every estimand is a false-minus-neutral difference against each condition's own oracle; the reverse stopping error is reported.
- **Utility and cost assumptions determine the stopping oracle.** One primary $(c, S, T)$ fixed and disclosed in advance; a prespecified sweep.
- **Model and oracle must have identical information.** The prompt is audited against the oracle's inputs; any gap would make the comparison meaningless.
- **Prior induction may not take.** Elicited priors seed the oracle; the instructed-versus-elicited gap is reported; surviving effects are reproduced with misleading-pilot induction.
- **Synthetic worlds may not generalize.** The design trades realism for an exact oracle on purpose, after study 4 showed that real data without an oracle cannot answer the question.
- **Prompt fragility.** About five equivalent formulations at the pilot stage, with prompt variant as a random effect.
- **Model-family heterogeneity.** Several families; the sign of the deviation is the cross-family criterion; models failing the competence gate are excluded rather than tuned.

- **World-selection overfitting.** Selection rules depend only on oracle geometry; thresholds come from the enumeration; the configuration is frozen before confirmatory runs; the rejection rate is logged.
- **Selective mechanism hunting after nulls.** The stop rule below, fixed in advance.

**What would count as success.** Opposition positive: do not add a mechanism; replicate on held-out worlds, run the prompt sweep, add model families, run the dose-response, reproduce with misleading-pilot induction. Stopping positive: it becomes the flagship; quantify closure rate, value destroyed, belief-state robustness, and cost/utility sensitivity. Both robust: the allocation extension below becomes the next study.

**What would falsify the hypothesis.** Our preferred story is weakened if false-prior models track same-prior optimal choice in opposition worlds, show little prior-specific excess closure, and those nulls hold across models and prompts, while the synthetic agents show the benchmark would have caught an effect of meaningful size at the power we ran. If that happens, we do not invent another confirmation-bias metric. The result becomes: much of the apparent confirmation bias in AI scientific agents disappears once the rational consequences of different priors are controlled, and some apparent persistence effects can be explained by ordinary Bayesian preservation of prior odds. That is a real paper, provided the benchmark is demonstrably sensitive.

*Stop rule.* If opposition and stopping are null under high neutral competence, sensitive synthetic-agent controls, multiple model families, and tight intervals, we stop searching for a positive confirmation-bias mechanism and write the robustness result.

## 10 Conditional Extension: Budget Allocation

Only if both primary studies are positive. The scientist has several open projects and one budget, and at each step chooses which project gets the next experiment. For two projects let $M = V_X - V_Y$ be the difference in rational marginal value; a rational agent switches near $M = 0$. If false-prior agents require substantially larger $M$ before returning to the question that threatens their belief, that is an implicit shadow cost on disconfirmatory inquiry. Deferred until earned.

The arc is short. Different priors rationally imply different behavior. So compare to a rational scientist with the same prior. Under that comparison, simple evidence-assimilation bias mostly disappears. So test whether the distortion has moved to the control of inquiry itself. The question is no longer whether an AI scientist can update when contradictory evidence arrives. It is whether it will keep doing the science that lets the contradiction arrive at all.

## References


D. Agarwal, B. P. Majumder, R. Adamson, M. Chakravorty, S. R. Gavireddy, A. Parashar, H. Surana, B. Dalvi Mishra, A. McCallum, A. Sabharwal, and P. Clark. AutoDiscovery: Open-ended scientific discovery via Bayesian surprise. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2507.00310.

K. Atwell, P. Heydari, A. Sicilia, and M. Alikhani. BASIL: Bayesian assessment of sycophancy in LLMs. *arXiv preprint arXiv:2508.16846*, 2025.

R. M. Batista and T. L. Griffiths. A rational analysis of the effects of sycophantic AI. *arXiv preprint arXiv:2602.14270*, 2026.

L. Bertolazzi, K. Tentori, and R. Bernardi. FalsifyBench: Evaluating inductive reasoning in LLMs with rule discovery games. *arXiv preprint arXiv:2606.04751*, 2026.

D. A. Boiko, R. MacKnight, B. Kline, and G. Gomes. Autonomous chemical research with large language models. *Nature*, 624 (7992):570–578, 2023. doi: 10.1038/s41586-023-06792-0.

K. Chaloner and I. Verdinelli. Bayesian experimental design: A review. *Statistical Science*, 10(3):273–304, 1995. doi: 10.1214/ss/1177009939.

C. Chen, M. Jörke, A. Goliński, M. Fedzechkina, G. Sapiro, S. Williamson, and N. Foti. LLMs are not (consistently) Bayesian: Quantifying internal (in)consistencies of LLMs' probabilistic beliefs. *arXiv preprint arXiv:2605.06915*, 2026.

Y. Chen, P. Piękos, M. Ostaszewski, F. Laakom, and J. Schmidhuber. PhysGym: Benchmarking LLMs in interactive physics discovery with controlled priors. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2025. arXiv:2507.15550.

C. Cheng and X. Huan. Optimal stopping for sequential Bayesian experimental design. *arXiv preprint arXiv:2509.21734*, 2025.

D. Choudhury, S. Williamson, A. Goliński, N. Miao, F. Bickford Smith, M. Kirchhof, Y. Zhang, and T. Rainforth. BED-LLM: Intelligent information gathering with LLMs and Bayesian experimental design. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2508.21184.

M. H. DeGroot. *Optimal Statistical Decisions*. McGraw-Hill, New York, 1970.

A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth. Deep adaptive design: Amortizing sequential Bayesian experimental design. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 3384–3395. PMLR, 2021.

K. Gandhi, M. Y. Li, L. Goodyear, A. Bhatia, L. Li, A. Bhaskar, M. Zaman, and N. D. Goodman. BoxingGym: Benchmarking progress in automated experimental design and model discovery. In *NeurIPS 2025 Workshop on Scaling Environments for Agents*, 2025. arXiv:2501.01540.

T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

G. Grand, V. Pepe, J. Andreas, and J. B. Tenenbaum. Shoot first, ask questions later? Building rational agents that explore and act like people. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2510.20886.

R. Gupta, R. Corona, J. Ge, E. Wang, D. Klein, T. Darrell, and D. M. Chan. Enough coin flips can make LLMs act Bayesian. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025a. arXiv:2503.04722.

R. Gupta, J. Hartford, and B. Liu. LLMs for Bayesian optimization in scientific domains: Are we there yet? In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025b. arXiv:2509.21403.

R. A. Howard. Information value theory. *IEEE Transactions on Systems Science and Cybernetics*, SSC-2(1):22–26, 1966. doi: 10.1109/TSSC.1966.300074.

P. Jansen, M.-A. Côté, T. Khot, E. Bransom, B. Dalvi Mishra, B. P. Majumder, O. Tafjord, and P. Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. arXiv:2406.06769.

A. R. Jhaveri, A. GX-Chen, I. Sucholutsky, and E. Choi. Failing to falsify: Evaluating and mitigating confirmation bias in language models. In *ICLR 2026 Workshop on From Human Cognition to AI Reasoning: Models, Methods, and Applications*, 2026. arXiv:2604.02485.

J. Klayman and Y.-W. Ha. Confirmation, disconfirmation, and information in hypothesis testing. *Psychological Review*, 94 (2):211–228, 1987. doi: 10.1037/0033-295X.94.2.211.

D. V. Lindley. On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4): 986–1005, 1956. doi: 10.1214/aoms/1177728069.

C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.

R. S. Nickerson. Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, 2(2):175–220, 1998. doi: 10.1037/1089-2680.2.2.175.

M. Oaksford and N. Chater. A rational analysis of the selection task as optimal data selection. *Psychological Review*, 101(4): 608–631, 1994.

A. Pal, T. Kitanovski, A. Liang, A. Potti, and M. Goldblum. Knowing what you know is not enough: Large language model confidences don't align with their actions. *arXiv preprint arXiv:2511.13240*, 2025.

T. Rainforth, A. Foster, D. R. Ivanova, and F. Bickford Smith. Modern Bayesian experimental design. *Statistical Science*, 39 (1):100–114, 2024. doi: 10.1214/23-STS915.

M. Ríos-García, N. Alampara, C. Gupta, I. Mandal, S. Mannan, A. A. Aghajani, N. M. A. Krishnan, and K. M. Jablonka. AI scientists produce results without reasoning scientifically. *arXiv preprint arXiv:2604.18805*, 2026.

M. Sclar, Y. Choi, Y. Tsvetkov, and A. Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.11324.

M. Sharma, M. Tong, T. Korbak, D. Duvenaud, et al. Towards understanding sycophancy in language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.13548.

K. Tian, E. Mitchell, A. Zhou, A. Sharma, R. Rafailov, H. Yao, C. Finn, and C. D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. arXiv:2305.14975.

A. Wald. *Sequential Analysis*. John Wiley & Sons, New York, 1947.

P. C. Wason. On the failure to eliminate hypotheses in a conceptual task. *Quarterly Journal of Experimental Psychology*, 12(3): 129–140, 1960. doi: 10.1080/17470216008416717.

J. Xie, K. Zhang, J. Chen, R. Lou, and Y. Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2305.13300.

M. Xiong, Z. Hu, X. Lu, Y. Li, J. Fu, J. He, and B. Hooi. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2306.13063.

Y. Yamada, R. T. Lange, C. Lu, S. Hu, C. Lu, J. Foerster, J. Clune, and D. Ha. The AI scientist-v2: Workshop-level automated scientific discovery via agentic tree search. *arXiv preprint arXiv:2504.08066*, 2025.

K. Yamin, J. Tang, S. Cortes-Gomez, A. Sharma, E. Horvitz, and B. Wilder. When agents say one thing and do another: Validating elicited beliefs from LLMs. *arXiv preprint arXiv:2602.06286*, 2026.